Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Java** and **Python** backend services, **search and retrieval** systems, storage and indexing, and production **LLM** workflows, with rigorous testing and validation of AI-generated output at scale.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed multi-tenant storage on Azure SQL and Blob Storage with row-level isolation, tuning **indexes and query** access patterns to cut dashboard latency from 1.2s to 350ms for 500+ users.
- Restructured fault-tolerant asynchronous **LLM orchestration** with FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a production LLM serving backend on AWS routing requests across GPT-4o, Claude, and Qwen-VL behind one switchable interface, handling **10K+ daily requests** for 150+ instructors with stable peak-load responses.
- Engineered concurrent batch processing with ThreadPoolExecutor and **validated model outputs** through OCR fallbacks and grading-consistency checks, sustaining 90%+ consistency across 10K+ submissions while cutting feedback turnaround 60%.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer in PyTorch for offline reinforcement learning, with end-to-end **data pipelines** processing 100K+ trajectory logs, improving long-horizon trajectory planning 18% in closed-loop simulation.
- Designed a spatial-temporal encoder using **multi-head attention** over modality-specific map and agent token embeddings, benchmarking against three baseline encoders and improving closed-loop planning stability by 15% in simulation.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Optimized large-scale **Java** and Python API integrations with pagination, field filtering, and **caching** of frequently requested data, improving response times 30% and shrinking payloads for low-bandwidth clients.
- Built object-oriented microservice modules with Singleton and Factory patterns, backed by 20+ **unit and integration** suites in Mockito and PyTest, and redesigned ETL workflows on AWS Glue boosting pipeline efficiency 68%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **Java Spring Boot** microservices with RESTful APIs and real-time WebSocket updates for a resource-management platform, synchronizing 2M+ records across distributed locations with 33% fewer sync conflicts.
- Automated Jenkins **CI/CD** with unit, integration, and end-to-end tests, Docker packaging, and zero-downtime rolling releases, cutting deployment time 50% and reaching 95%+ successful build validation across release runs.

Skills

Languages: Java, Python, C++, SQL, JavaScript, TypeScript, Bash
Search & Retrieval: FAISS, Pinecone, Vector Search, Semantic Search, Embeddings, Indexing, Query Optimization, Retrieval-Augmented Generation
Backend & Distributed: Spring Boot, FastAPI, Flask, REST APIs, Microservices, Async Processing, Caching, Distributed Systems, Concurrency
Data & Storage: PostgreSQL, Azure SQL, SQL Server, MongoDB, Blob Storage, AWS S3
AI & Practices: LLM Systems, PyTorch, Transformers, Unit Testing, PyTest, Mockito, CI/CD, Docker, Agile, Git

Projects

Hybrid Lexical-Semantic Retrieval Service | *Java, Lucene, FAISS, Python, Docker, JUnit*

- Designed a **Java** search service combining BM25 inverted-index retrieval with FAISS vector search, merging results through reciprocal rank fusion and evaluating **relevance** offline with recall and NDCG on labeled queries.

Segmented Inverted Index Library | *Java, Write-Ahead Log, Concurrency, JUnit, JMH*

- Designed an embeddable Java **storage library** with immutable index segments, a **crash-safe write-ahead log**, background segment merging, and lock-free concurrent readers, validated through unit tests and microbenchmarks.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*

- Built a **semantic retrieval** platform with transformer embeddings, FAISS, and Pinecone over 50K+ documents, adding incremental indexing and tuning to cut retrieval latency 40% while sustaining 95% query accuracy.